Measures of Disease Occurrence - Part 1

APHS 20203 • Epidemiology & Biostatistics

Course Learning objectives

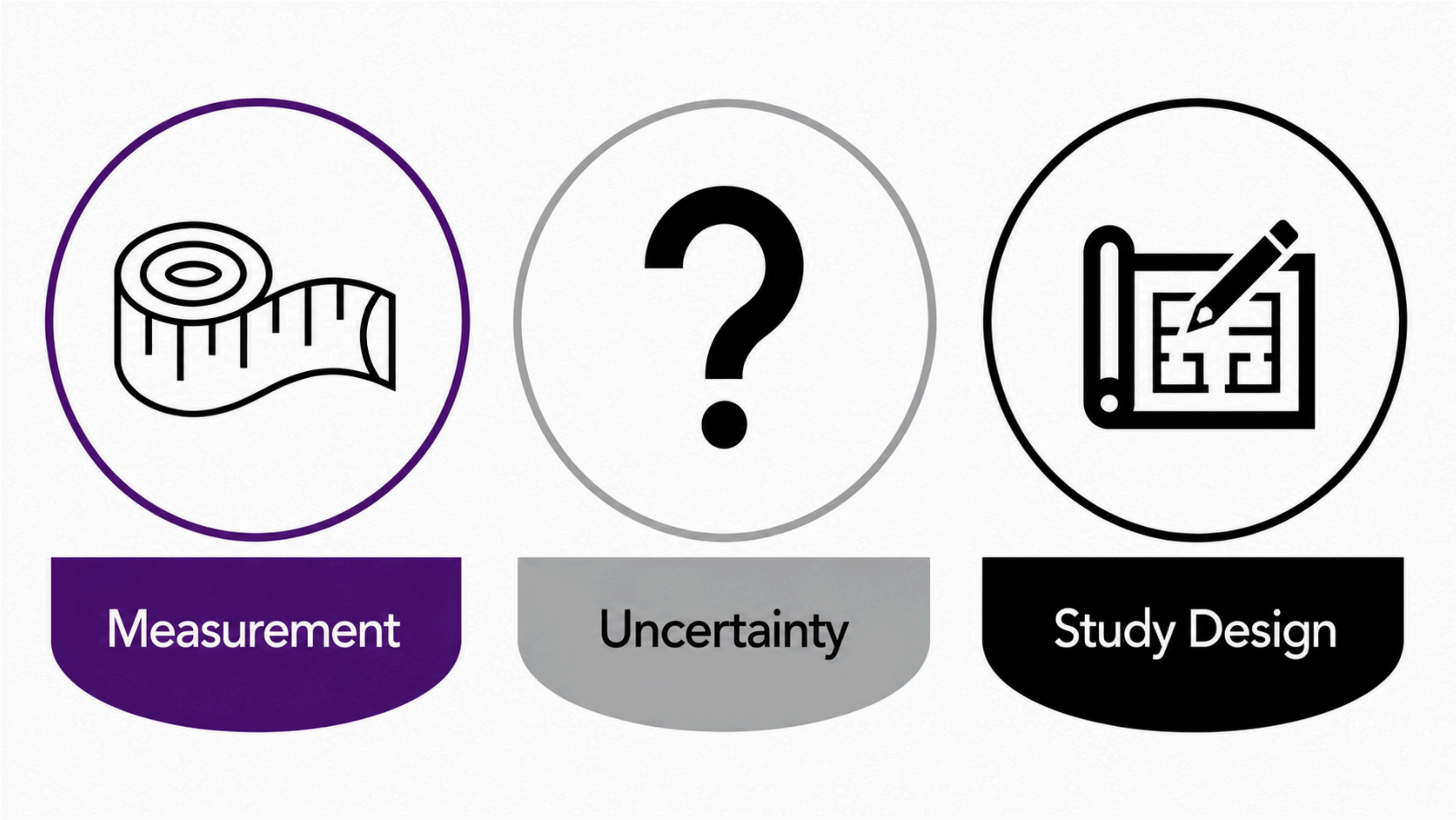
1. Define epidemiology and key associated terminology, especially confounding.
2. Distinguish among association, prediction, and causation.
3. Explain the Rothman model of sufficient-component causes (RMSCC) and the concept of multifactorial causation.
4. Explain the limitations of anecdotal evidence in drawing epidemiologic conclusions.
5. Explain why disseminating risk information is sometimes necessary but rarely sufficient to improve population health.

Lecture Learning objectives

By the end of this lesson, you'll be able to:

1. Define counts and rates.
2. Distinguish among crude, specific, and adjusted rates.
3. Compute and interpret prevalence and incidence.

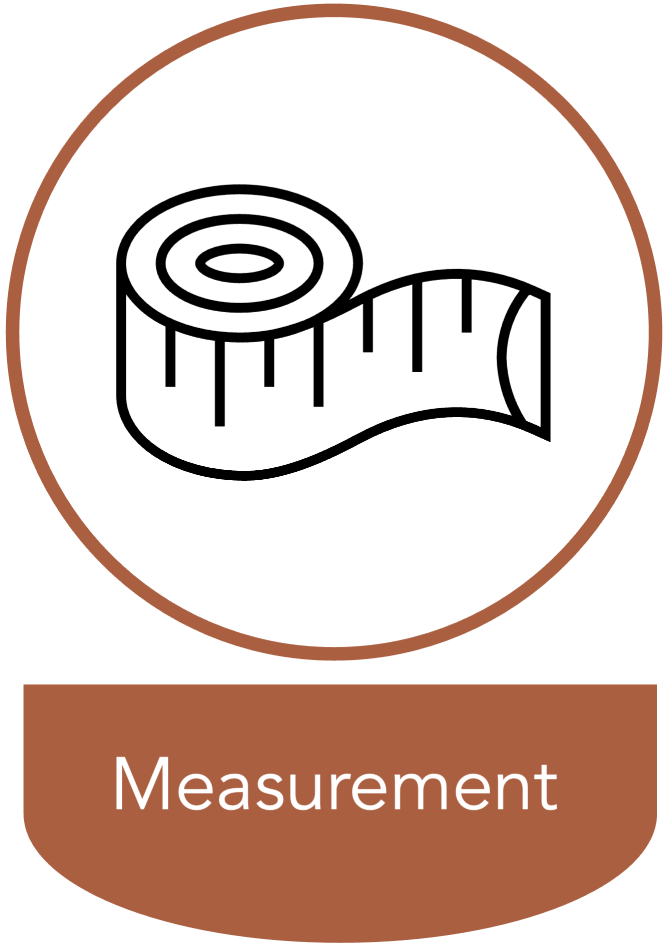
Measurement, Uncertainty, and Study Design



What is Epidemiology?

> Epidemiology is concerned with the distribution and determinants of health and diseases, morbidity, injuries, disability, and mortality in populations. Epidemiologic studies are applied to the control of health problems in populations (Quinlan 2025).

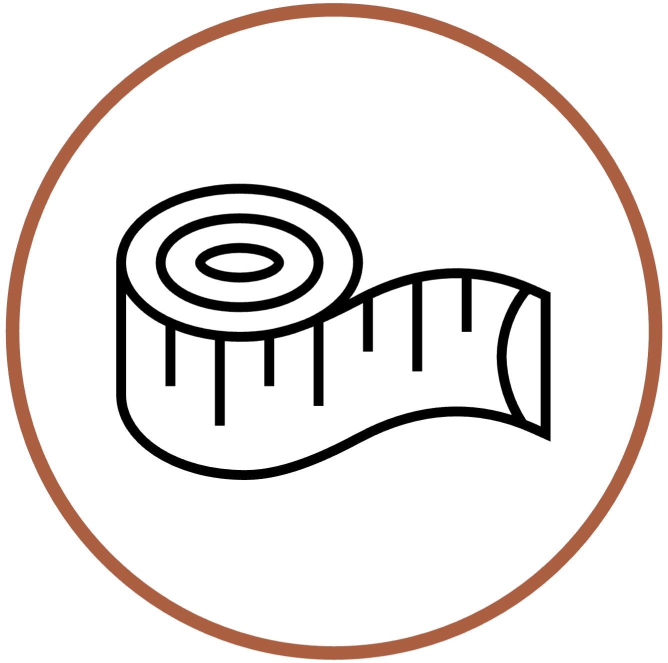
Measurement Makes Characteristics Observable



> Epidemiology is concerned with the **distribution** and determinants of health and diseases, morbidity, injuries, disability, and mortality in populations. Epidemiologic studies are applied to the control of health problems in populations (Quinlan 2025).

- Observe and assign values to relevant characteristics.
- Look for patterns among those values.

What Does It Mean to Observe?



Measurement

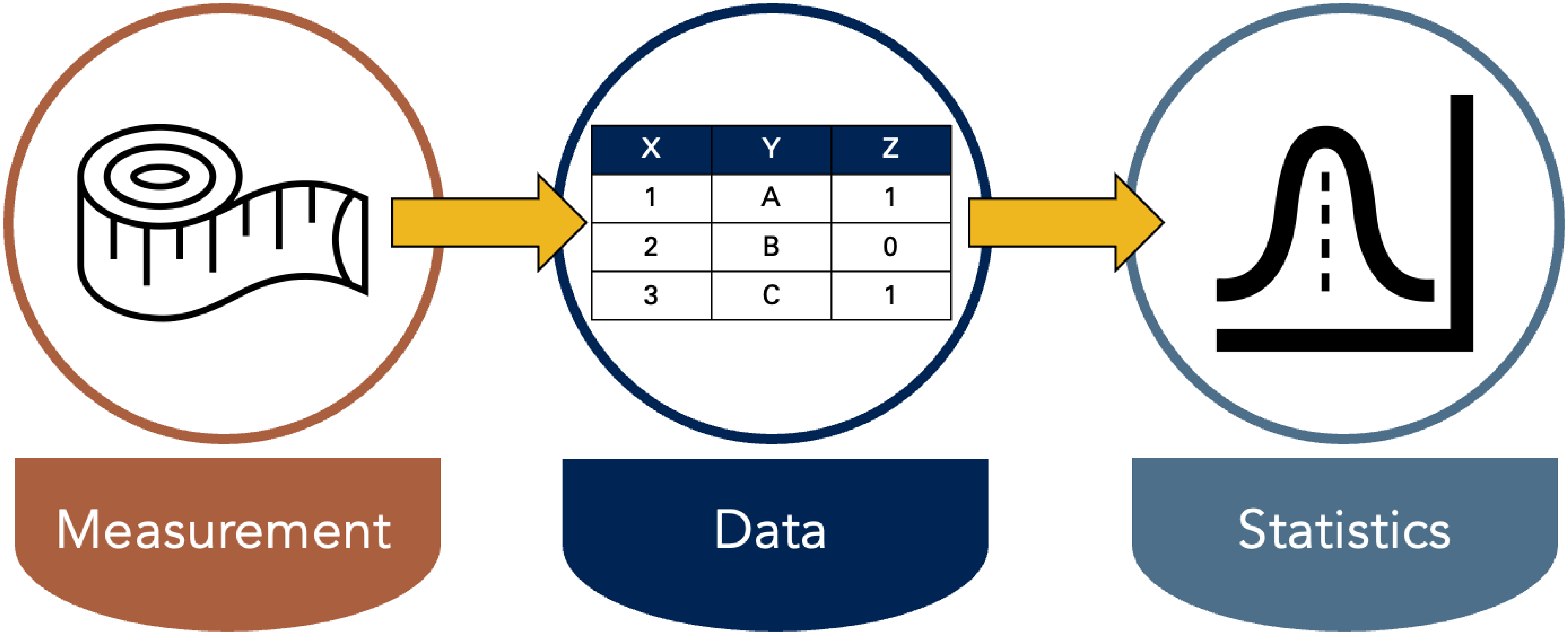
Epidemiology is concerned with the distribution and **determinants** of health and diseases, morbidity, injuries, disability, and mortality in populations. Epidemiologic studies are applied to the control of health problems in populations (Quinlan 2025).

- Observe and assign values to relevant characteristics.
- Look for patterns among those values.

OBSERVE

“To be aware of” or “have knowledge about.”

Measurement Produces Data; Statistics Finds Patterns



Description, Prediction, and Causation

DESCRIPTION	PREDICTION	CAUSATION

Description Does Not Necessarily Seek Associations

- Description examines distributions of single variables: middle, spread, shape, count, and proportion.
- It supports resource management and planning.
- Examples:
  - How many ventilators are available in Texas?
  - What is the average age of people living in Florida?
  - How much time elapses, on average, between exposure to a pathogen and the occurrence of disease symptoms?

Description Can Also Compare Distributions

- Sometimes description does look for associations.
- It may compare distributions of single variables within levels of another variable.
- Examples:
  - Are more ventilators available in Texas or New York?
  - Are people older, on average, in Florida or Pennsylvania?
  - Is symptom onset quicker, on average, for cholera or *E. coli*?

Measures of Occurrence Support Description

- Counts
- Incidence
- Prevalence
- Odds

These measures can be useful on their own and for hypothesis generation.

Sources of Uncertainty: Few Checklists

1. Few checklists

Sources of Uncertainty: Add Statistical Uncertainty

1. Few checklists
2. Statistical uncertainty

Sources of Uncertainty: Add Causal Uncertainty

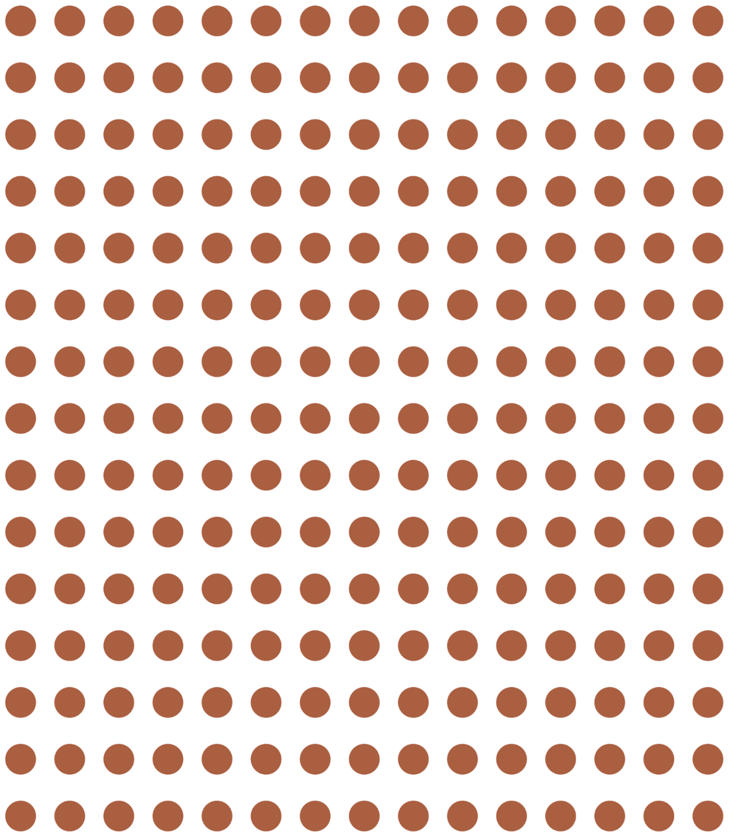
1. Few checklists
2. Statistical uncertainty
3. Causal uncertainty

Occurrence Is Defined Within Populations

> Epidemiology is concerned with the distribution and determinants of health and diseases, morbidity, injuries, disability, and mortality in **populations**. Epidemiologic studies are applied to the control of health problems in populations (Quinlan 2025).

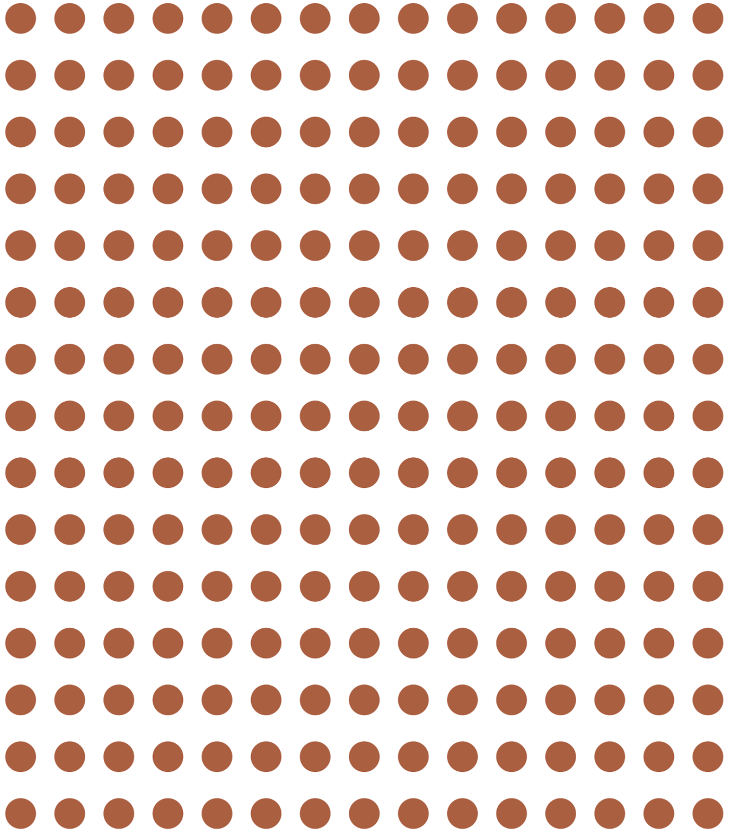
Population

> The simplest definition of a **population** is a group of people who share characteristics or meet criteria that define membership in the population (Lash et al. 2021).

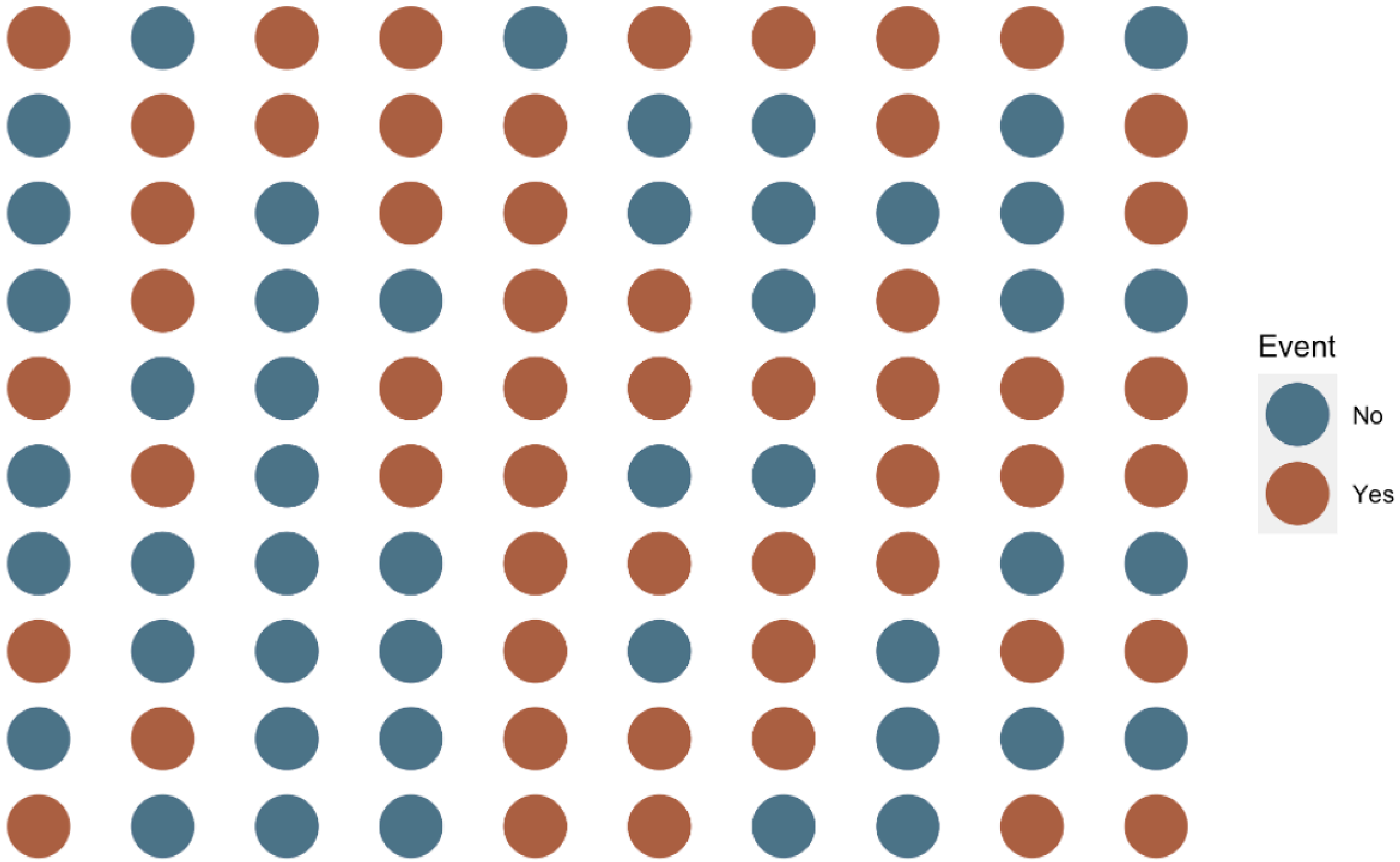


A Population Also Has a Time Period

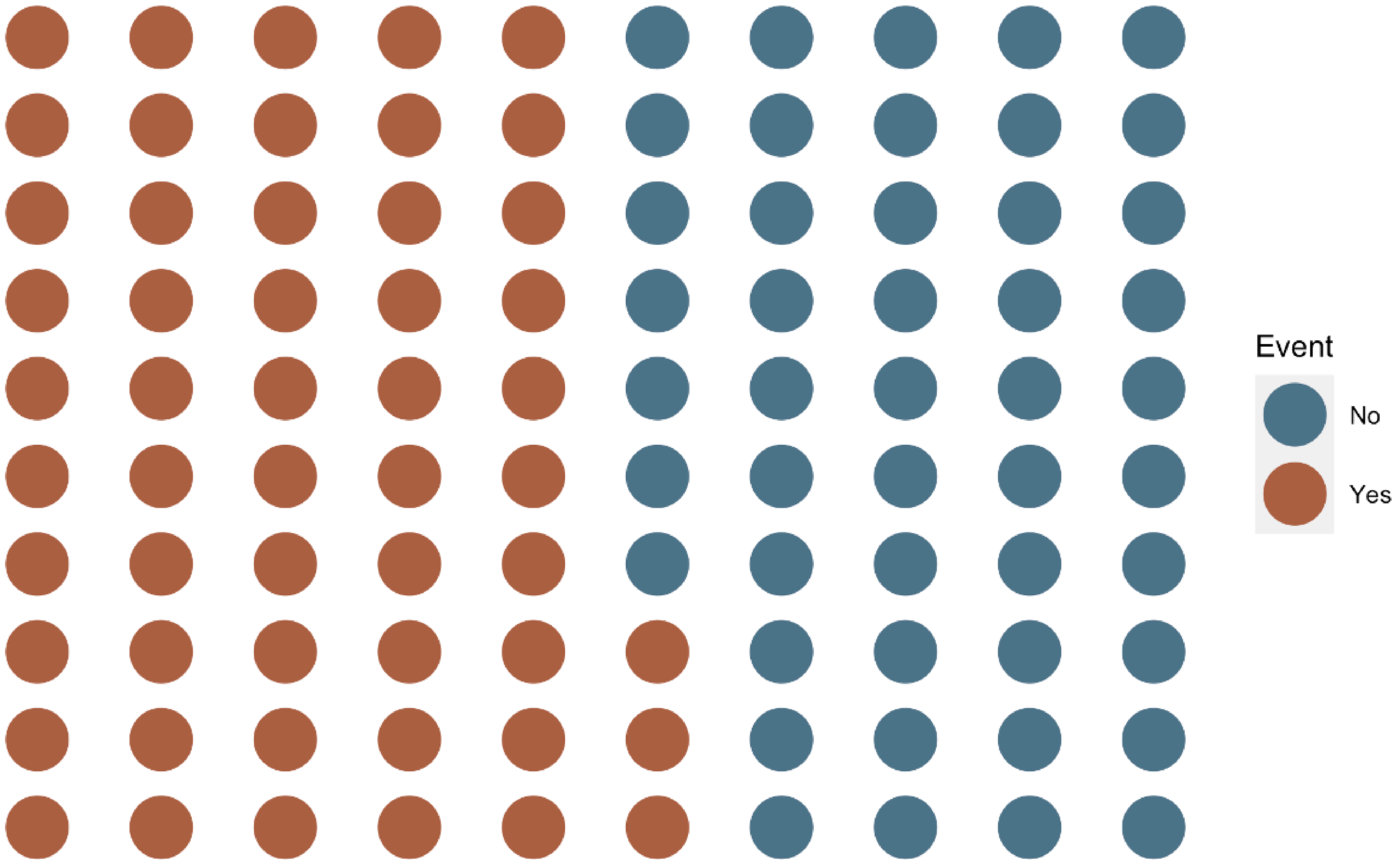
> The simplest definition of a **population** is a group of people who share characteristics or meet criteria that define membership in the population [**during a defined time period**] (Lash et al. 2021).



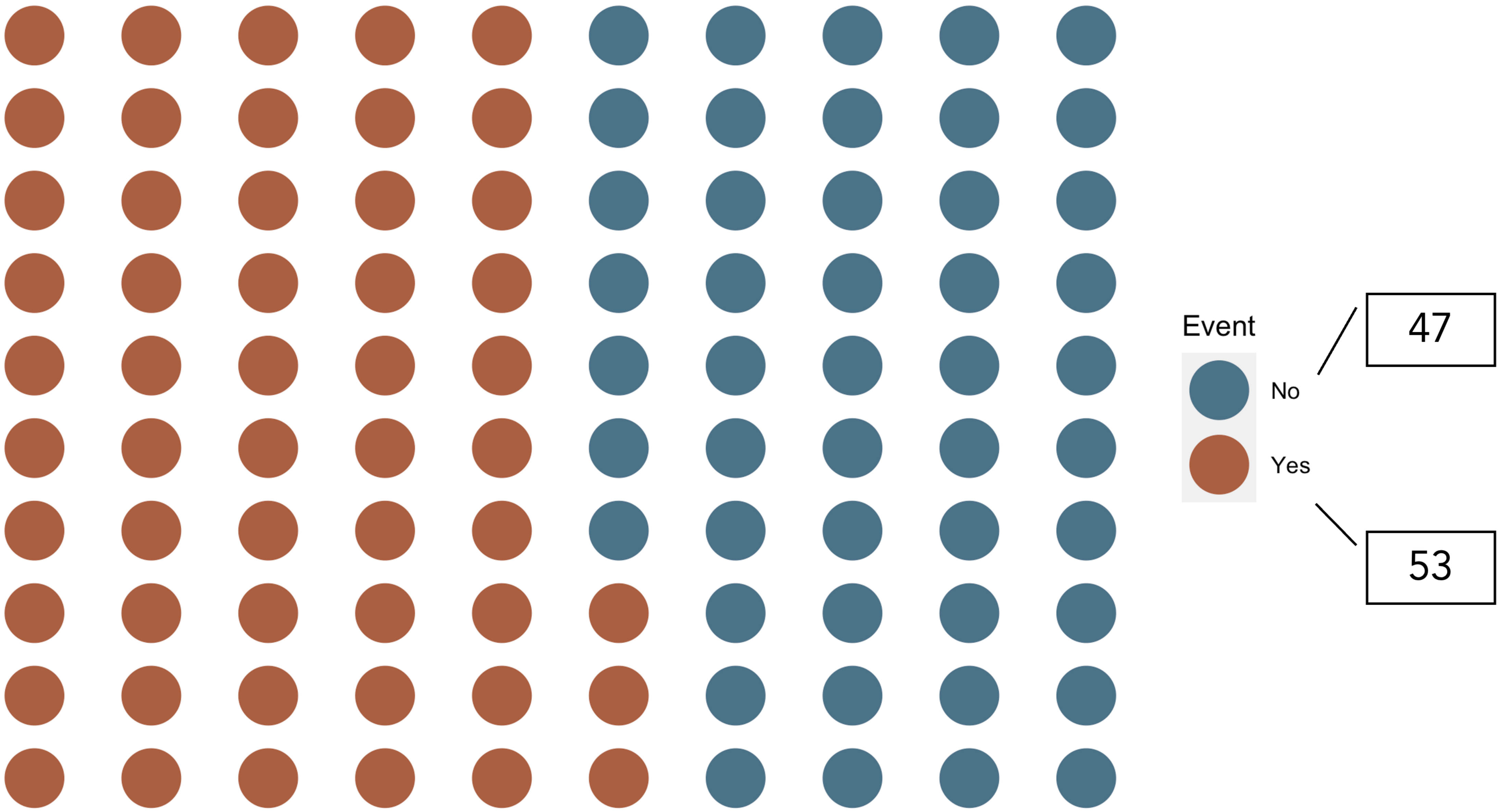
Counts (n): Events in a Population



Counts (n): Group the Event Categories



Counts (n): Reveal the Totals



The Same Count Can Tell Different Stories

Imagine that **10 students have mono today** in each of two dorms.

	Dorm A	Dorm B
Students with mono	10	10
Total residents	20	500
Percentage with mono	50%	2%

The count helps us plan care. The **denominator** shows how common mono is.

A Ratio Compares Two Quantities

A **ratio** divides one quantity by another.

In a hypothetical clinic survey, **15 patients use e-cigarettes and 5 do not**.

$$\frac{\text{Users}}{\text{Nonusers}} = \frac{15}{5} = 3$$

There are **3 users for every 1 nonuser**: a ratio of **3:1**.

The denominator does not have to include the numerator.

A Proportion Compares a Part With the Whole

For the same **15 users and 5 nonusers**, there are **20 patients in all**.

$$\text{Proportion who use} = \frac{\text{Users}}{\text{All patients}} = \frac{15}{20} = 0.75$$

- Everyone counted in the numerator is also in the denominator.
- A proportion ranges from **0 to 1**.
- Here, **three-fourths of the patients** use e-cigarettes.

A Percentage Expresses a Proportion per 100

$$\text{Percentage who use} = \frac{15}{20} \times 100\% = 75\%$$

0.75 and 75% describe the same share of patients.

- Multiply a proportion by 100 to express it as a percentage.
- Divide a percentage by 100 to recover the proportion: $75/100 = 0.75$.
- "75%" means **75 out of every 100**, even when the actual group has 20 people.

A Rate Describes How Quickly Events Occur

A **rate** includes time in its denominator.

For an incidence rate, we divide new cases by **person-time at risk**.

$$\frac{6 \text{ new cases}}{120 \text{ person-years}} = 5 \text{ cases per } 100 \text{ person-years}$$

This describes the pace of new cases. It does **not** mean that 5% became ill.

Prevalence Describes Existing Conditions

- Prevalence provides information about the presence or absence of a condition of interest, such as a disease, exposure, or characteristic, in a population.
- It describes the extent to which the condition is present in a population during a given time frame.
- Asking about the prevalence of diabetes means asking how many, or more likely what proportion, of people currently living in the population have diabetes.

Source: (Westreich 2019)

Prevalence Count

- A count of people in a population with the condition of interest during a given time frame.
- As few as 0 people and as many as all people can be living with the condition.
- Range: 0 to infinity.

Source: (Westreich 2019)

Prevalence Proportion

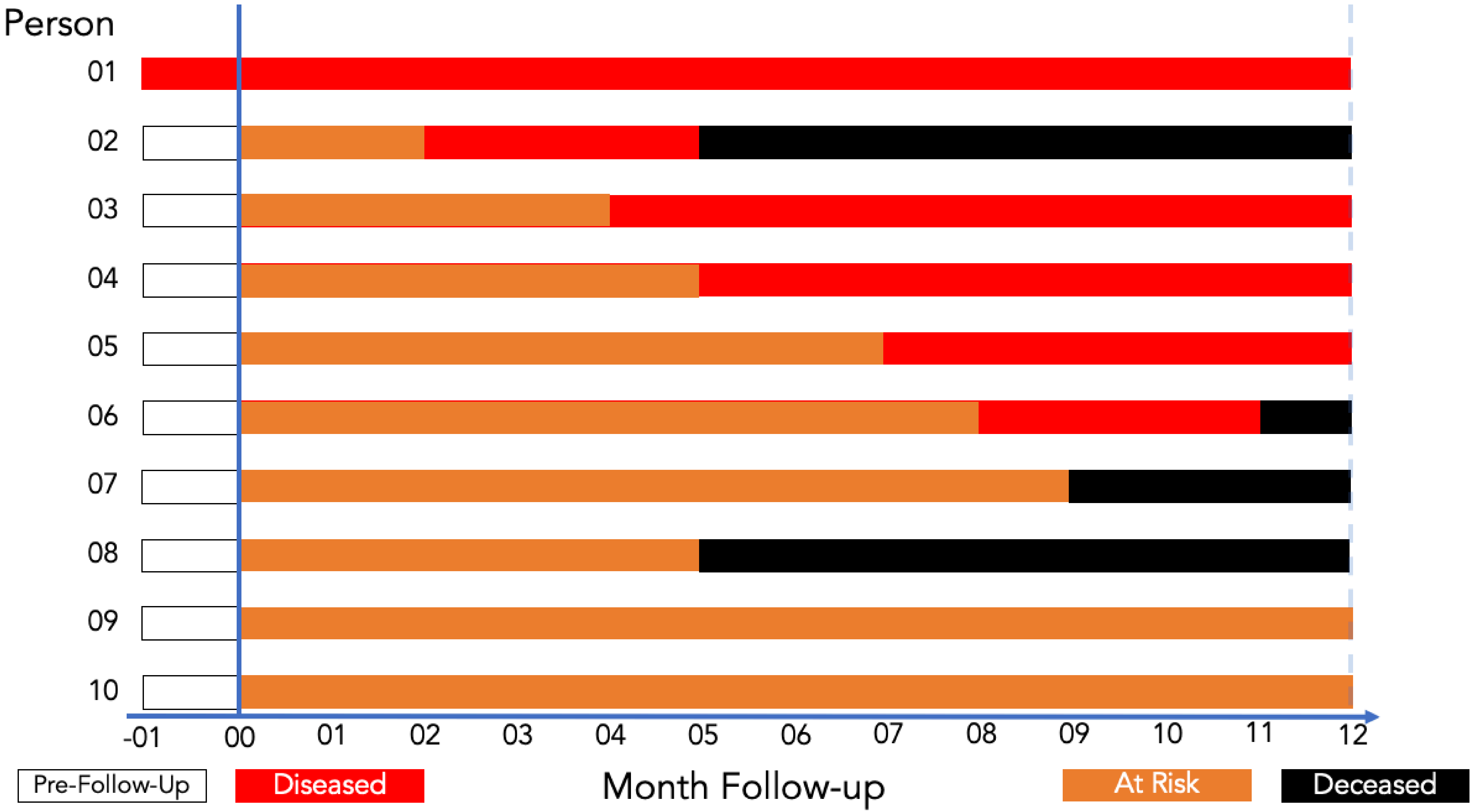
- The proportion or percentage of the population that presently has the condition of interest or presently has a history of the condition during a specified period, such as a history of cancer in the past 10 years.
- Since none, some, or all population members can have the condition or a history of it, prevalence proportion ranges from 0 to 1, much like a probability.

Source: (Westreich 2019)

Prevalence Proportion: Formula

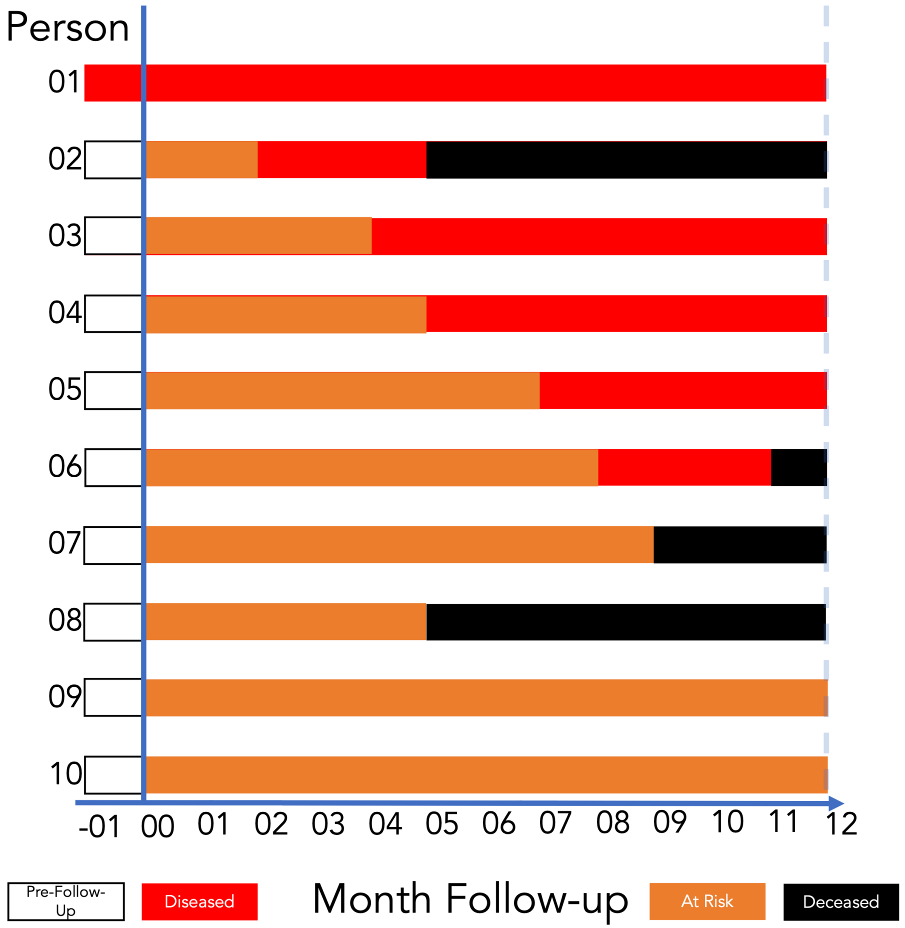
$$\text{Prevalence proportion} = \frac{\text{Count with the condition}}{\text{Living members of the population}}$$

Follow-Up Experience for 10 People



Source: Lash TL, VanderWeel TJ, Haneuse S, Rothman KJ. *Modern Epidemiology*. fourth. Wolters Kluwer; 2021.

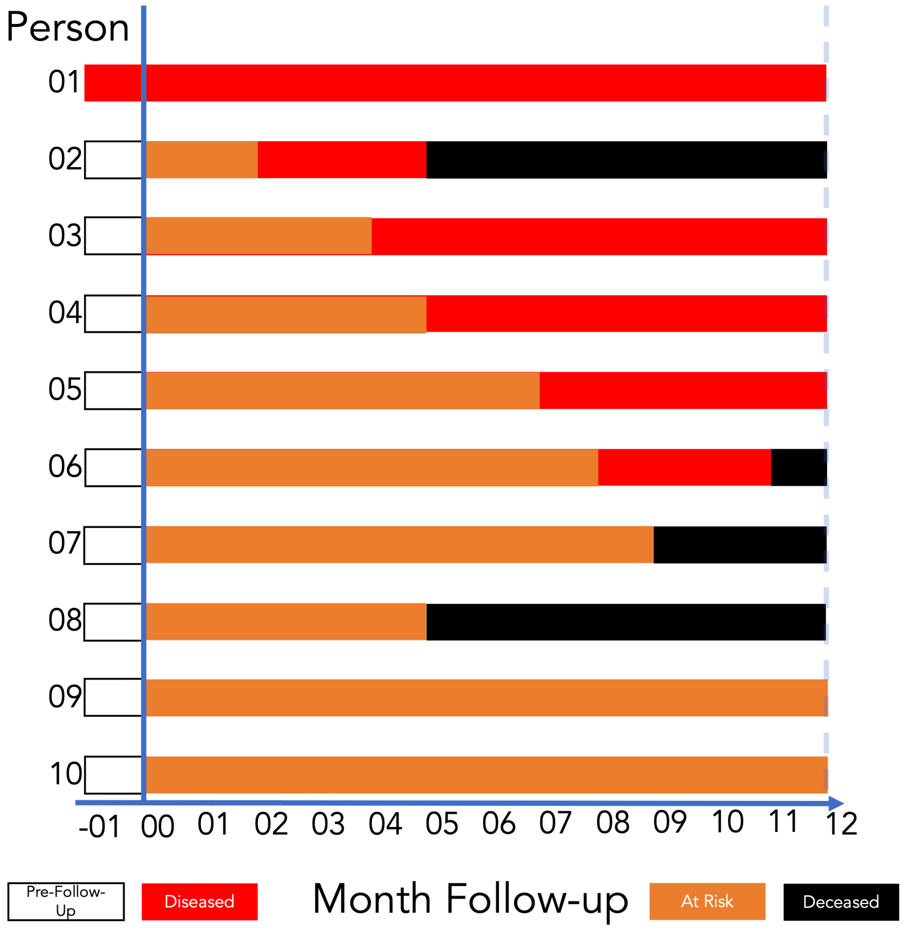
Prevalence Proportion at Month 2



$$\frac{\text{Count with the condition}}{\text{Living members of the population}}$$

$$= \frac{2}{10} = 0.20 = 20\%$$

Interpreting Prevalence Proportion



Among the members of the population, the prevalence of disease in month 2 was **0.20**.

Said another way, **20%** of the population had disease in month 2.

Point Prevalence and Period Prevalence

- **Point prevalence:** prevalence of a condition of interest at a single point in time.
- **Period prevalence:** prevalence of a condition of interest over a period of time.
- As the duration of the period shrinks, period prevalence converges to point prevalence.
- There is no hard-and-fast rule for what window is sufficiently short to count as point prevalence.

Source: (Westreich 2019)

Odds Compare People With and Without a Condition

In a hypothetical survey conducted today, **20 students have a headache** and **80 do not**.

$$\text{Odds of a headache} = \frac{\text{Students with a headache}}{\text{Students without a headache}} = \frac{20}{80} = \frac{1}{4} = 0.25$$

The odds are **1:4**: for every **1 student with a headache**, there are **4 without one**.

The Outcome You Name Goes on Top

To calculate the **odds of an event**:

$$\text{Odds of event} = \frac{\text{Number with the event}}{\text{Number without the event}}$$

To calculate the **odds of a non-event**, reverse the comparison:

$$\text{Odds of non-event} = \frac{\text{Number without the event}}{\text{Number with the event}}$$

For the headache example:

$$\text{Odds of headache} = \frac{20}{80} = 0.25 \qquad \text{Odds of no headache} = \frac{80}{20} = 4$$

The two odds are **reciprocals**: $0.25 = 1/4$ and $4 = 4/1$.

Rewrite 0.25 as Equivalent Odds

A decimal odds is a ratio with **1** in the denominator:

$$0.25 = \frac{0.25}{1} = 0.25 : 1$$

First find the number that turns 0.25 into 1:

$$1 \div 0.25 = 4$$

Then multiply both parts by 4:

$$(0.25 \times 4) : (1 \times 4) = \mathbf{1 : 4}$$

Multiplying both parts by the same number gives equivalent odds:

$$0.25 : 1 = 1 : 4 = 2 : 8 = 5 : 20 = 20 : 80 = 25 : 100$$

Each expression says the number **with** a headache is one-fourth the number **without** one.

Probability and Odds Use Different Denominators

The same survey has **20 students with a headache** and **80 without one**.

Measure	Calculation	Interpretation
Probability	$20/(20+80)=0.20$	20 out of every 100 students have a headache.
Odds	$20/80=0.25$	1 student has a headache for every 4 without one.

Probability compares the part to the whole. Odds compare one part to a different part.

Reading Odds Below, Equal to, and Above 1

Three hypothetical groups surveyed on the same day:

With a headache	Without a headache	Odds	What the odds tell us
20	80	$0.25 = 1 : 4$	Fewer have a headache than do not.
50	50	$1 = 1 : 1$	Equal numbers have and do not have a headache.
80	20	$4 = 4 : 1$	More have a headache than do not.

Odds can be **greater than 1**. A probability cannot.

Move Between Probability and Odds

Let p be the probability, written as a proportion between 0 and 1.

$$\text{Odds} = \frac{p}{1-p} \qquad p = \frac{\text{Odds}}{1+\text{Odds}}$$

- Probability **0.20** gives odds $0.20/0.80 = \mathbf{0.25}$.
- Odds **4:1** give probability $4/(4+1) = \mathbf{0.80}$, or **80%**.

For odds written as **a:b**, the probability is **a/(a+b)**.

Prevalence Odds

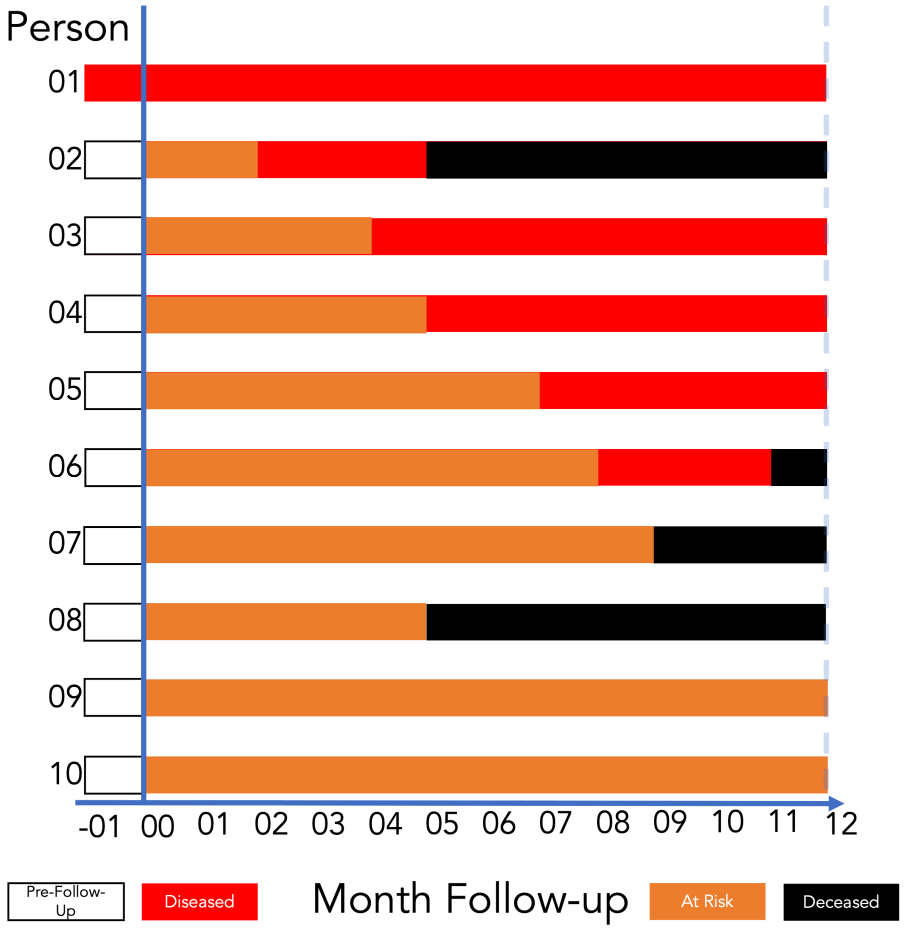
- Prevalence odds is a simple function of prevalence proportion, just as odds in general is a simple function of probability.
- It is typically reported for convenience or desirable statistical properties.
- Prevalence odds is prevalence proportion divided by 1 minus prevalence proportion.
- Range: 0 to infinity.

Source: (Westreich 2019)

Prevalence Odds: Formula

$$\text{Prevalence odds} = \frac{\text{Prevalence proportion}}{1 - \text{Prevalence proportion}}$$

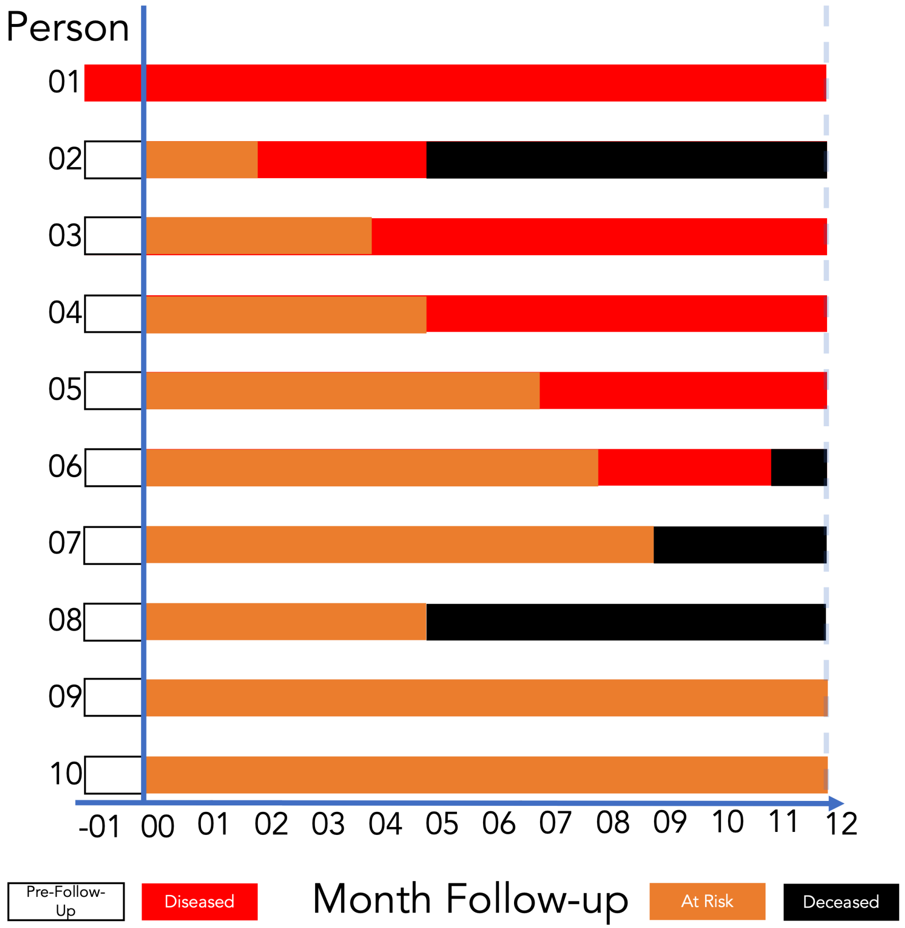
Prevalence Odds at Month 2



$$\frac{\text{Prevalence proportion}}{1 - \text{Prevalence proportion}}$$

$$\text{Prevalence odds} = \frac{0.20}{1 - 0.20} = 0.25$$

Interpreting Prevalence Odds



Among the members of the population, the odds of disease in month 2 were **0.25**.

For every person who had disease at month 2, there were four people who did not.

Why Learn to Calculate Odds?

- **Describe a condition:** compare how many people have it with how many do not.
- **Compare groups:** an *odds ratio* compares the odds in one group with the odds in another.
- **Understand statistical results:** some models use odds to describe how outcomes vary with other characteristics.

Understanding odds helps us translate those results into statements about people.

An Odds Ratio Compares Two Groups

In two hypothetical groups surveyed today:

Group	With a headache	Without a headache	Odds
A	60	40	$60/40 = 1.5$
B	50	50	$50/50 = 1.0$

$$\text{Odds ratio} = \frac{\text{Odds in A}}{\text{Odds in B}} = \frac{1.5}{1.0} = 1.5$$

Group A has **1.5 times the odds** of a headache compared with Group B today.

Questions?

What questions do you have?

Respond here

References

Lash, Timothy L., Tyler J. VanderWeele, Sebastien Haneuse, and Kenneth J. Rothman. 2021. *Modern Epidemiology*. 4th ed. Wolters Kluwer.

Quinlan, Scott. 2025. *Friis' Epidemiology 101*. 3rd ed. Jones & Bartlett Learning.

Westreich, Daniel. 2019. *Epidemiology by Design: A Causal Approach to the Health Sciences*. Oxford University Press.